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ASSIGNMENT

LEVEL 05

COMP50063

Data Analytics

AI2521/CYS2521/CNC2521/AIK2521

Hand Out Date: As Specified in SAIS

Hand In Date: As Specified in SAIS

INSTRUCTION TO CANDIDATES

1. Students are advised to underpin their answers with the use of references (cited using the Harvard Referencing Style).
2. Late submission will be awarded zero (0) unless extenuating circumstances (EC) are upheld.
3. Cases of plagiarism will be penalized.
4. The assignment should be submitted as a softcopy: The softcopy of the written assignment, artefact and the presentation should be submitted to given link in the LMS.
5. We draw your attention to the Academic Conduct Procedure to encourage good academic practice.

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Part 1 - Scope, Define and Explain

1. Introduction & Context

The retail industry generates tons of transactional data from customer purchases to feedback although organizations use data analytics to turn that info into useful business decisions. This project uses a synthetic online retail data set of 1,000 transactions to show how analytics could help make better business decision-making.

Using Microsoft Power BI, which enabled data preparation, transformation, modelling, visualization of the data to build dashboards. The study focuses on identifying buying trends, understanding customer habits, sales performance, and spotting any data quality issues. (Kaggle, n.d.)

2. Problem Statement

Retail businesses gather large amounts of transaction data, however, have a hard time turning it into useful insights. Poor data quality issues slow down analysis and negative impact on business choices. This project aims to change complex retail transaction data into reliable data to support sales analysis, understand customer behaviors, and boost smart decision-making.

3. Why Is It a Problem?

There were several problems with the dataset:

- Missing values in customer review scores.
- Missing values in gender information.
- There were also huge volumes of transaction records that needed organizing.
- Hard to identify revenue trends without the right visual tools.
- Raw data wasn't set up for direct business reporting.

These problems might cause incorrect findings and decrease trust in analytic outcomes.

4. Proposed Solution

The proposed solution was to set up a full data analytics process using Power BI.

The workflow included:

- Importing and validating data.
- Doing ETL processes.
- Cleaning the data.

- Making calculated measures.
- Exploring the data.
- Developing dashboards.
- Generating insights.

This method transforms raw retail data into useful visual reports to help with decisions.

5. Aim and Objectives

Aim

To analyze online retail transactions data with Power BI and get useful business insights, I need to process, visualize, and evaluate the data effectively.

Objectives

- To clean the retail dataset.
- To apply ETL techniques to improve data quality.
- To analyze customer buying trends and sales performance.
- To create interactive dashboards to help with informed decision-making.

The project succeeds when spotting and fixing data issues.

6. Success Criteria

The project will be considered successful if:

- Spot and fix data problems.
- Create insightful visuals.
- Draw useful business conclusions from the data.
- The dashboard supports collaborative investigation of data.
- Learning outcomes are demonstrated through critical reflection and evaluation.

7. Big Data Definition and Approach

Big Data deals with massive sets of information that move fast, vary widely, are usually reliable, and super valuable. However, this specific dataset only includes 1,000 records. This shows lots of real-world analytical traits, especially in data quality and management.

Power BI was selected because it provides everything for ETL processing, data modeling, nice visualizations and dashboards. This platform changes raw data into useful insights effectively and follows business intelligence standards well.

Part 2: Reflection on Experiences

8. Dataset Selection

First, Select Synthetic Online Retail Data from Kaggle for this project since it offers realistic e-commerce transactions. This helps with expressive business analysis. The dataset has 1,000 entries across five product categories, from March 2024 to March 2025. It covers Electronics, Fashion, Sports and Outdoors, Books and Stationery, and Home and Living.

The dataset was also selected because it contains data quality issues like missing values in the review score and gender columns. These problems were perfect for showing ETL processes and data cleaning techniques for this assignment. (Kaggle, n.d.)

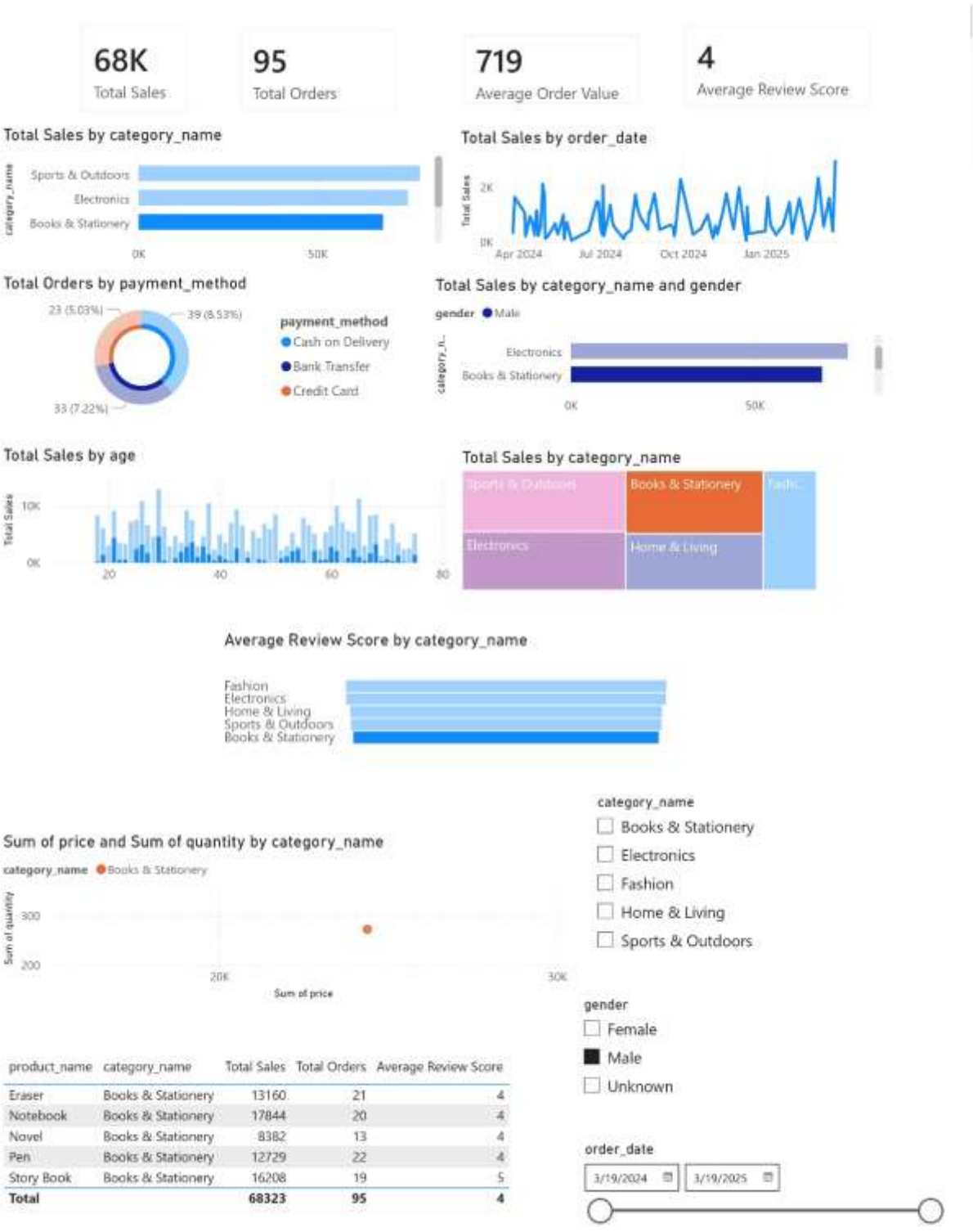
9. Big Data Fit

Big Data is defined by Volume, Velocity, Variety, Veracity and Value In this the selected dataset includes numerical, categorical, and date data types. The missing values highlight an issue with Veracity, which means data quality problems need fixing before analyzing the data effectively. The dataset suggestions business Value by showing sales trends, customer behavior, and how products perform. Even though it doesn't cover Volume and Velocity like large scale Big Data would, it still works great as a case study to practice Big Data analytics techniques.

10. Visualization and Approach Justification

Microsoft Power BI was picked for this analytical because it offers everything needed for ETL, data modeling, DAX calculations, and interactive dashboard creation in one place. We made a simple, one-page dashboard with ten visualizations that answer the project objectives

The dashboard uses KPI cards to show Total Sales, Total Orders, Average Order Value, Average Review Score. Other charts include a clustered bar chart for Sales by Category, a line chart for Total Sales by order date, and a donut chart for Orders by Payment Method. There's also a stacked bar chart displaying Sales by Category and Gender, and a column chart for Total Sales by age. And a tree map for Total Sales by category, a funnel chart for Average Review Score by category name, and a scatter chart looking at Sum of price and Sum of quantity by category, Table Visual for Product Summary. To make things more interactive, we included slicers for Category, Gender, and Order Date. That helps users filter the data however they want.



product_name	category_name	Total Sales	Total Orders	Average Review Score
Eraser	Books & Stationery	13160	21	4
Notebook	Books & Stationery	17844	20	4
Novel	Books & Stationery	8382	13	4
Pen	Books & Stationery	12729	22	4
Story Book	Books & Stationery	16208	19	5
Total		**68323**	**95**	**4**

Figure 1 - Dashboard

11. ETL Issues

In Power BI Desktop, the ETL process happened entirely in Power Query. After importing the dataset, we found some data quality issues and fixed them in just seven transformation steps.

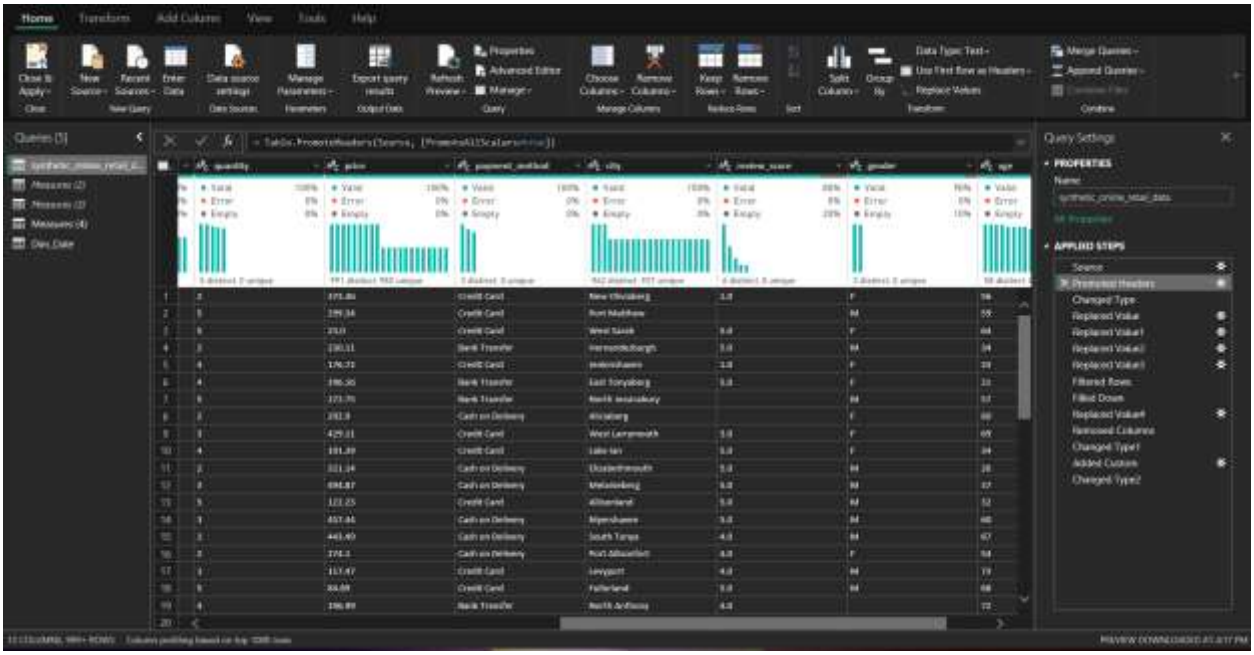


Figure 2 - Dataset Import — Before Cleaning

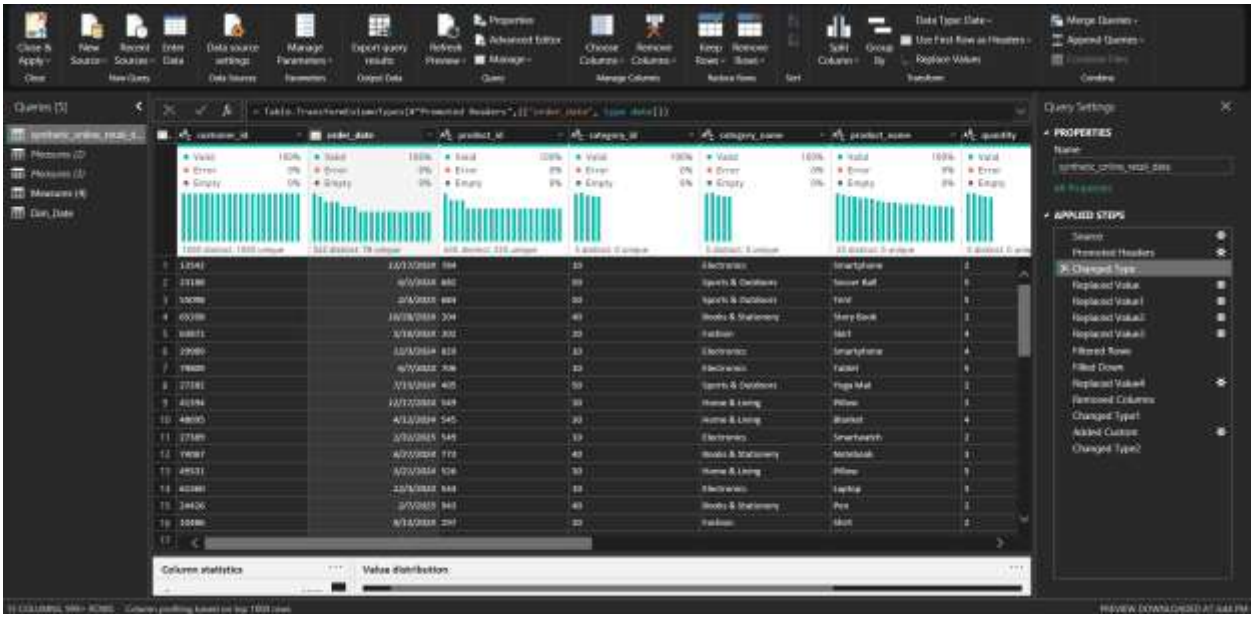


Figure 3 - Changed data type to Date

- Changed the order_date column from text to Date type to enable DAX time intelligence functions and fix monthly groupings in charts.

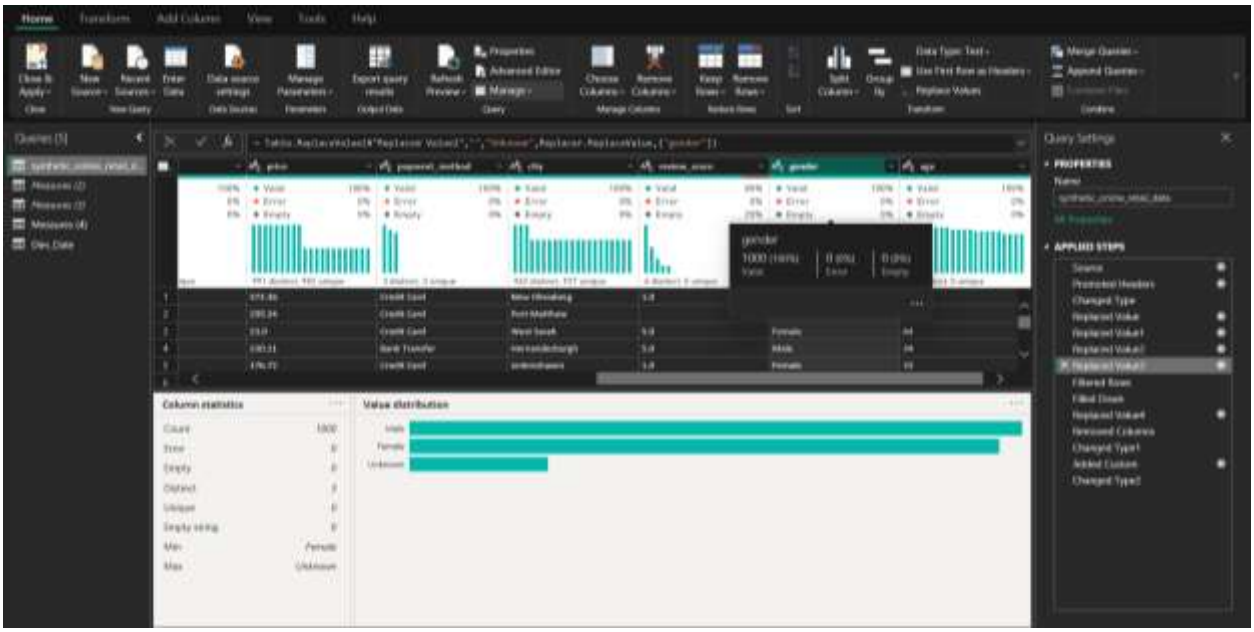


Figure 4 - Replaced value with Male & Female

- Also, F and M for gender were changed out with Female and Male using Replace Values, making the visuals easier to read. For missing gender info there were nulls. They filled in with Unknown to keep all records complete

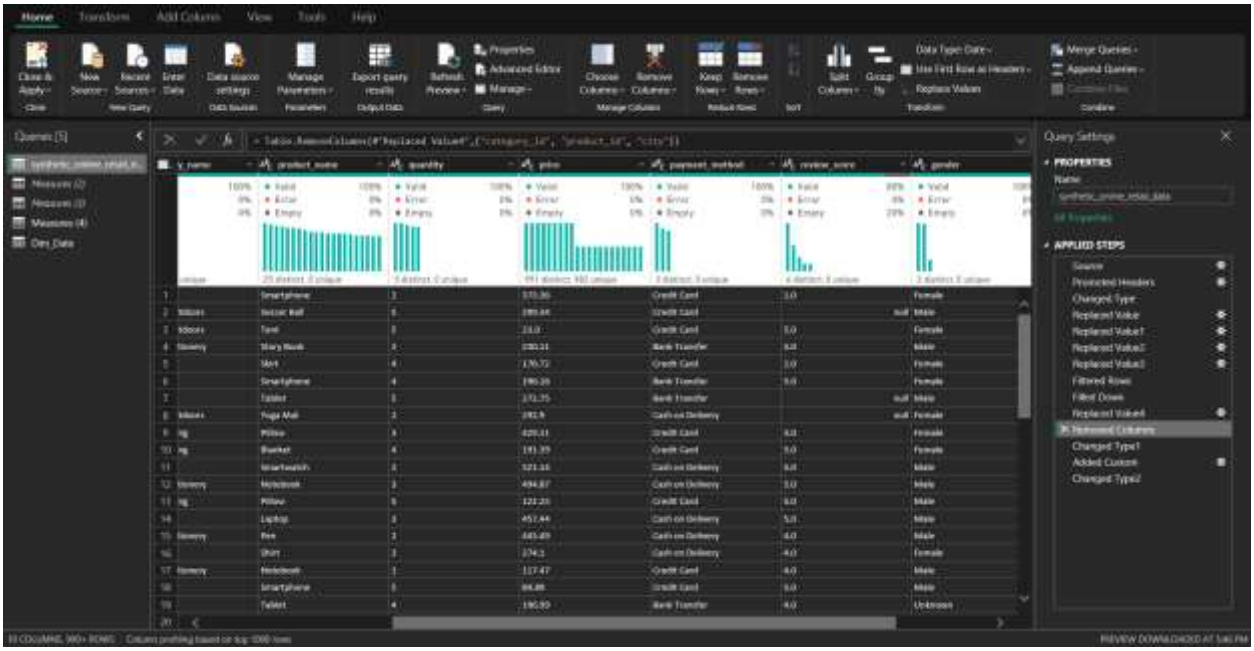


Figure 5 - Remove unnecessary columns

- Three extra columns of category_id, product_id, and city were removed. Because it's not providing grouping value.

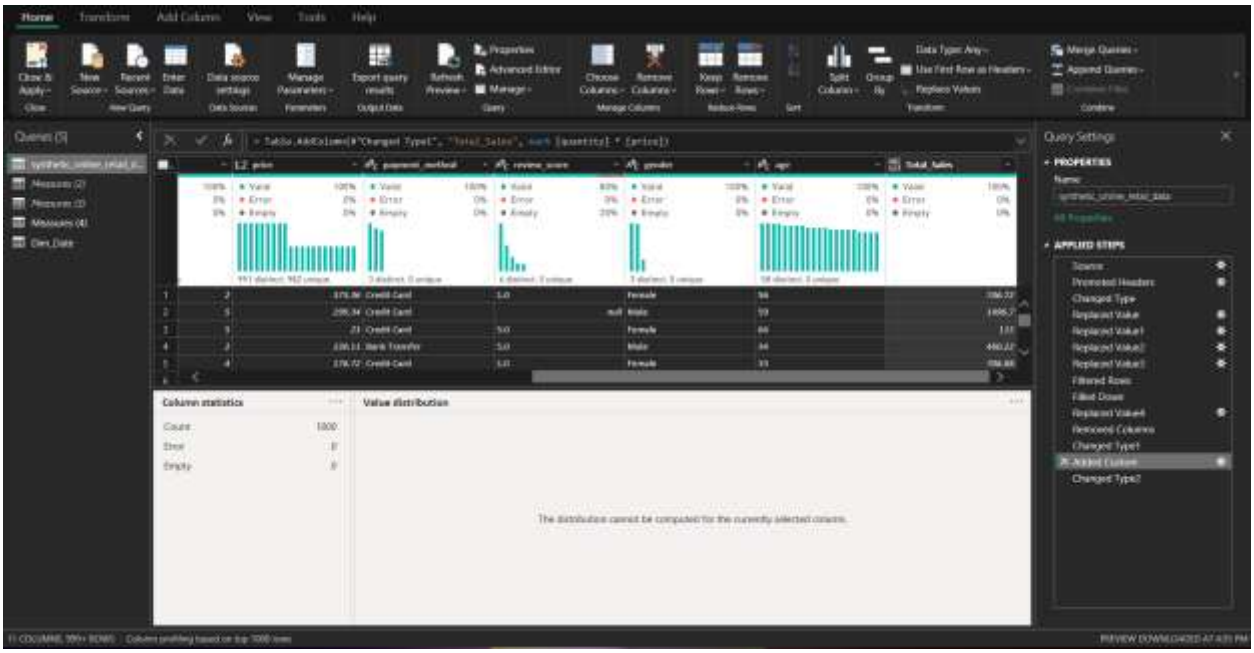


Figure 6 - Create custom collumn

- Then, added a Total_Sales custom column by multiplying quantity by price since there wasn't a revenue field. Finally, made all column names easier to understand.

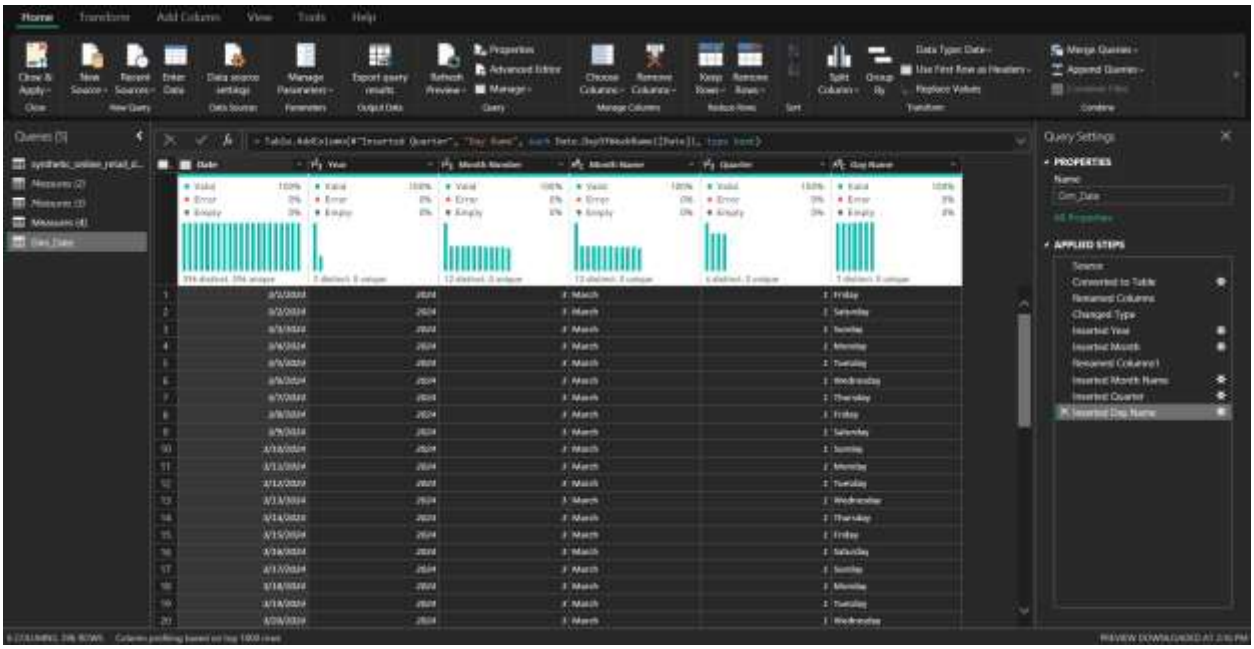


Figure 7 - Created dimension table

- I made a Date dimension table called Dim_Date using Power Query. I created a date range from March 2024 to March 2025 and added columns for Year, Month Number, Month Name, Quarter, and Day Name. I did this using the Date functions in the Add Column tab.

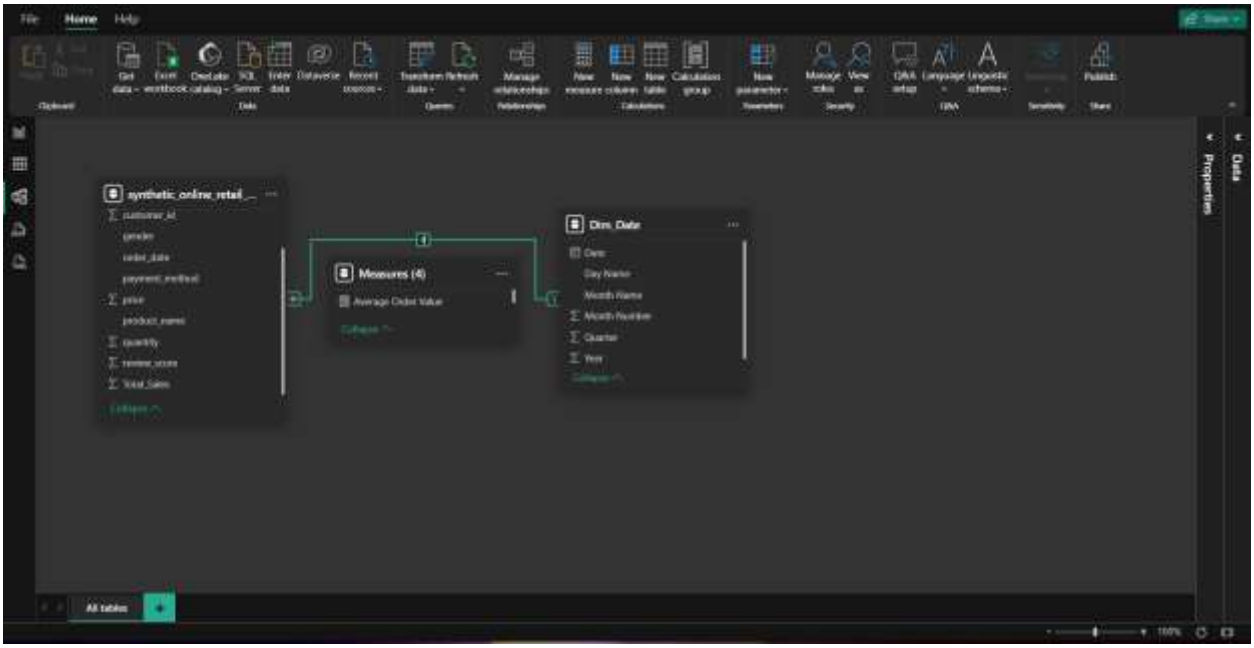


Figure 8 - Data model - Star schema

- Then, in Model View, I linked this table to the main fact table with a one-to-many relationship based on Order Date. So, I designed a star schema data model.

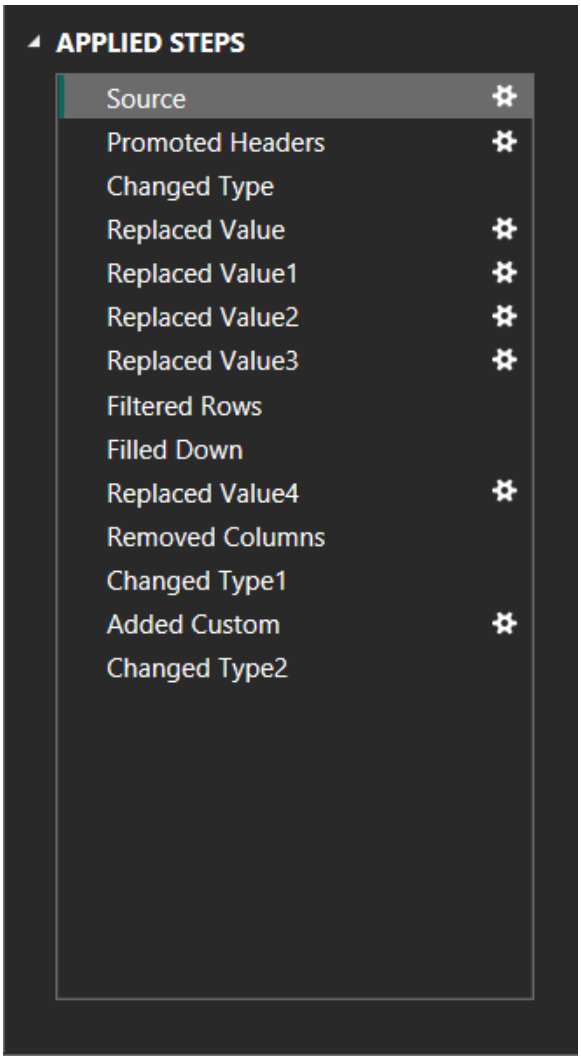


Figure 9 - Applied Steps Panel

12. Learning Outcomes Reflection

The first thing looked at was talking about and evaluating techniques for processing Big Data. For this project, I checked out null retention, Unknown replacement, and column removal. We picked the right technique for each problem created on how it affected analysis accuracy. This helped us make smart choices about which methods to use.

The second learning outcome involved finding and assessing the top ways to handle Big Data analytics. We went with Power BI since it supports the whole analytics process. I chose analytical techniques that turned raw data into useful insights on sales trends, customer behavior, and product performance.

The third learning outcome was to present well-reasoned arguments for why we picked certain methods. Whenever we cleaned data, transformed it, or came up with designs for visualizations, backed up decisions with reasons connected to the data set and the goals of the project. I decided on chart types based on what made sense for the data, bar charts for comparing categories, line charts for showing trends, and KPI cards for highlighting key metrics.

Final learning outcome was, created a Power BI dashboard to showcase this Big Data research. It includes everything from gathering big data to cleaning it up, modeling, creating DAX measures, and ending with interactive visualizations for insights.

13. Evaluation of Power BI

Power BI was chosen since it provides everything for data cleaning, transformation, modeling, and visualization in one platform. Tools like Power Query, DAX, and interactive dashboards were useful during the analysis.

But some challenges were faced. Date formatting caused errors, and creating relationships and DAX measures required a lot of extra studying and problem solving. Power BI proved its effectiveness in analyzing the data and generating valuable business insights. (Microsoft, n.d.)

14. References

Kaggle. (n.d.). Retrieved from https://www.kaggle.com/datasets/ertugrulesol/online-retail-data

Microsoft. (n.d.). Retrieved from https://www.microsoft.com/en-us/power-platform/products/power-bi